Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 10 Solutions

Problem 10.1

I have two coins, one fair (F) and one biased (B). Suppose I keep picking a coin and tossing it. You get to observe whether the coin I tossed landed heads (H) or tails (T). Your goal is to guess the sequence of coins I used from the observed coin toss results.

For the first toss, I choose either the fair coin or the biased coin uniformly at random. I prefer not to switch coins too often: after each toss, I use the same coin with probability 3/4 and switch coins with probability 1/4, independently of all other tosses. The fair coin is equally likely to land heads or tails. The biased coin lands heads with probability 1/4 and tails with probability 3/4.

Determine the most likely sequence of coin choices as you observe THT.

Solution: Since we have three observations $(Y_1, Y_2, Y_3) = (\texttt{THT})$ the HMM has three hidden states X_1, X_2, X_3 and we seek

$$(\hat{x}_1, \hat{x}_2, \hat{x}_3) = \arg\max_{(x_1,x_2,x_3)} p_{X_1,X_2,X_3|Y_1,Y_2,Y_3}(x_1, x_2, x_3|\texttt{THT}).$$

We fold the observations in by defining appropriate node potentials

$$\tilde{\phi}_n(x_n) = \begin{cases} p_{X_1}(x_1)\, p_{Y_1|X_1}(y_1|x_1) & n = 1 \\ p_{Y_n|X_n}(y_n|x_n) & n = 2, 3. \end{cases}$$

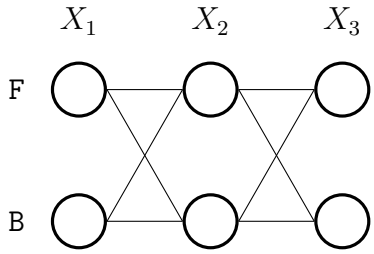
Based on the problem setup, we obtain

$$\begin{aligned} \tilde{\phi}_1(\texttt{F}) &= p_{X_1}(\texttt{F})\, p_{Y_1|X_1}(\texttt{T}|\texttt{F}) = \frac{1}{4}, & \tilde{\phi}_1(\texttt{B}) &= p_{X_1}(\texttt{B})\, p_{Y_1|X_1}(\texttt{T}|\texttt{B}) = \frac{3}{8}, \\ \tilde{\phi}_2(\texttt{F}) &= p_{Y_2|X_2}(\texttt{H}|\texttt{F}) = \frac{1}{2}, & \tilde{\phi}_2(\texttt{B}) &= p_{Y_2|X_2}(\texttt{H}|\texttt{B}) = \frac{1}{4}, \\ \tilde{\phi}_3(\texttt{F}) &= p_{Y_3|X_3}(\texttt{T}|\texttt{F}) = \frac{1}{2}, & \tilde{\phi}_3(\texttt{B}) &= p_{Y_3|X_3}(\texttt{T}|\texttt{B}) = \frac{3}{4}. \end{aligned}$$

The edge potential is the same for both edges:

$$\phi_{n,n+1}(x_n, x_{n+1}) = p_{X_{n+1}|X_n}(x_{n+1}|x_n) = \begin{cases} 3/4 & x_{n+1} = x_n \\ 1/4 & x_{n+1} \neq x_n, \end{cases} \qquad n = 1, 2.$$

We employ the Viterbi algorithm to find the most likely sequence (x_1, x_2, x_3). The trellis diagram below defines the graph in which we will compute the shortest path:



We will use logarithms base-2 and define $\gamma \triangleq \log_2 3 \approx 1.6$. As with the forward-backward algorithm, we use the vector form that stacks the values of the messages for all values of x. We adopt the convention that the first component of the vector corresponds to $x = \mathtt{F}$ and the second component corresponds to $x = \mathtt{B}$. Using this convention, we have

$$\log \tilde{\phi}_1(\cdot) = -\begin{bmatrix} 2 \\ 3-\gamma \end{bmatrix} \qquad \log \tilde{\phi}_2(\cdot) = -\begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \log \tilde{\phi}_3(\cdot) = -\begin{bmatrix} 1 \\ 2-\gamma \end{bmatrix}$$

$$\log \phi_{n,n+1}(x_n, x_{n+1}) = -\begin{cases} 2-\gamma & x_{n+1} = x_n, \\ 2 & x_{n+1} \neq x_n. \end{cases}$$

The min-sum algorithm messages are:

$$m_{1\to 2}(\cdot) \triangleq \min_{x_1}\big[-\log \tilde{\phi}_1(x_1) - \log \phi_{1,2}(x_1, \cdot)\big],$$
$$m_{n\to(n+1)}(\cdot) \triangleq \min_{x_n}\big[m_{(n-1)\to n}(x_n) - \log \tilde{\phi}_n(x_n) - \log \phi_{n,n+1}(x_n, \cdot)\big], \quad n = 2, 3,$$

with the trace-back messages $t_{(n+1)\to n}$ holding the values of x_n that achieve the minimum in the expressions above.

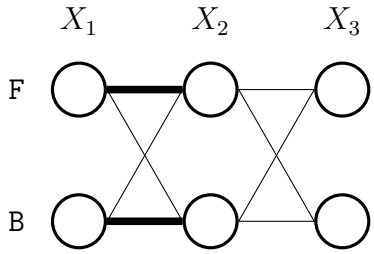
With a slight abuse of notation, we can express the first message as follow:

$$\begin{aligned} m_{1\to 2}(\cdot) &\triangleq \min_{x_1}\big[-\log \tilde{\phi}_1(x_1) - \log \phi_{1,2}(x_1, \cdot)\big] \\ &= \min\Bigg\{\underbrace{\begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 2-\gamma \\ 2 \end{bmatrix}}_{x_1=\mathtt{F}}, \underbrace{\begin{bmatrix} 3-\gamma \\ 3-\gamma \end{bmatrix} + \begin{bmatrix} 2 \\ 2-\gamma \end{bmatrix}}_{x_1=\mathtt{B}}\Bigg\} \\ &= \min\Bigg\{\underbrace{\begin{bmatrix} 4-\gamma \\ 4 \end{bmatrix}}_{x_1=\mathtt{F}}, \underbrace{\begin{bmatrix} 5-\gamma \\ 5-2\gamma \end{bmatrix}}_{x_1=\mathtt{B}}\Bigg\} = \begin{bmatrix} 4-\gamma \\ 5-2\gamma \end{bmatrix}, \end{aligned}$$

where the minimum is taken separately for each value of x_2 (row). The corresponding trace-back message is

$$t_{2\to 1}(\cdot) = \begin{bmatrix} \mathtt{F} \\ \mathtt{B} \end{bmatrix},$$

which indicates that for $x_2 = \mathtt{F}$ (first row), we chose $x_1 = \mathtt{F}$, and for $x_2 = \mathtt{B}$ (second row), we chose $x_1 = \mathtt{B}$. This is illustrated in the following first-step trellis diagram:



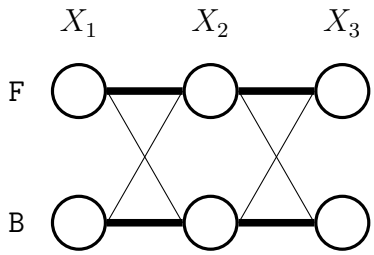
The second message is computed similarly:

$$m_{2\to 3}(\cdot) = \min\Bigg\{\underbrace{\begin{bmatrix}4-\gamma\\4-\gamma\end{bmatrix}+\begin{bmatrix}1\\1\end{bmatrix}+\begin{bmatrix}2-\gamma\\2\end{bmatrix}}_{x_2=\mathtt{F}},\underbrace{\begin{bmatrix}5-2\gamma\\5-2\gamma\end{bmatrix}+\begin{bmatrix}2\\2\end{bmatrix}+\begin{bmatrix}2\\2-\gamma\end{bmatrix}}_{x_2=\mathtt{B}}\Bigg\}$$
$$= \min\Bigg\{\underbrace{\begin{bmatrix}7-2\gamma\\7-\gamma\end{bmatrix}}_{x_2=\mathtt{F}},\underbrace{\begin{bmatrix}9-2\gamma\\9-3\gamma\end{bmatrix}}_{x_2=\mathtt{B}}\Bigg\} = \begin{bmatrix}7-2\gamma\\9-3\gamma\end{bmatrix},$$

and

$$t_{3\to 2}(\cdot) = \begin{bmatrix}\mathtt{F}\\\mathtt{B}\end{bmatrix}.$$

An illustration of the trace-back is:



We can now compute

$$\hat{x}_3(\mathtt{THT}) = \arg\min_{x_3}\big[m_{2\to 3}(x_3) - \log\tilde{\phi}_3(x_3)\big]$$
$$= \arg\min\begin{bmatrix}7-2\gamma+1\\9-3\gamma+2-\gamma\end{bmatrix} = \arg\min\begin{bmatrix}8-2\gamma\\11-4\gamma\end{bmatrix} = \mathtt{B},$$

where we computed the minimum over the elements of the vector. This in turn leads to a trace-back sequence

$$\hat{x}_2(\mathtt{THT}) = t_{3\to 2}(\hat{x}_3(\mathtt{THT})) = t_{3\to 2}(\mathtt{B}) = \mathtt{B},$$
$$\hat{x}_1(\mathtt{THT}) = t_{2\to 1}(\hat{x}_2(\mathtt{THT})) = t_{2\to 1}(\mathtt{B}) = \mathtt{B}.$$

Hence, the most likely sequence of coins is BBB.

As an interesting aside, we can also compute the most likely configuration after we observe `T` followed by `H`:

$$\begin{aligned}\hat{x}_2(\texttt{TH}) &= \arg\min_{x_2}\big[m_{1\to 2}(x_2) - \log\tilde{\phi}_2(x_2)\big] \\ &= \arg\min\begin{bmatrix}4-\gamma+1\\5-2\gamma+2\end{bmatrix} = \arg\min\begin{bmatrix}5-\gamma\\7-2\gamma\end{bmatrix} = \texttt{F}, \\ \hat{x}_1(\texttt{TH}) &= t_{2\to 1}(\hat{x}_2(\texttt{TH})) = t_{2\to 1}(\texttt{F}) = \texttt{F}.\end{aligned}$$

and

$$\hat{x}_1(\texttt{T}) = \arg\min_{x_1}\big[-\log\tilde{\phi}_1(x_1)\big] = \arg\min\begin{bmatrix}2\\3-\gamma\end{bmatrix} = \texttt{B}.$$

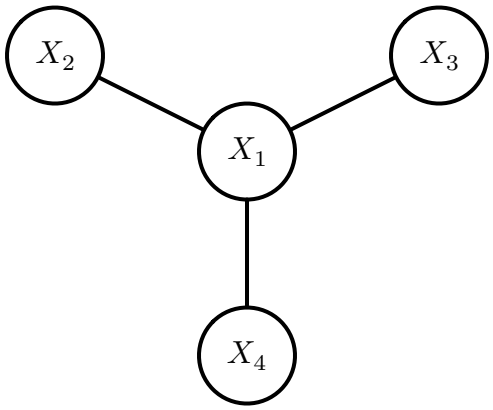
So if we observe tails (`T`) on the first toss, we conclude that the biased coin (`B`) was flipped (this is consistent with the MAP estimate—can you show that?—we can also verify that if we were to observe heads on the first toss, then we would have concluded that a fair coin was tossed). Now, if we observe two coin tosses that result in `TH`, the model of transitions favors the same coin used in both tosses, which yields the most likely configuration of `FF`. Observing `THT` again changes the most likely configuration to `BBB`.

Note that if we had modeled the choice of coins as independent of all other tosses (rather than a Markov chain), the most likely configuration would have been `BFB`.

Try running the Viterbi algorithm on the observations `HHT` and note how the transition model interacts with the evidence provided by the observations.

Problem 10.2

Let $X_1, X_2, X_3, X_4 \in \mathcal{X} = \{0, 1\}$ be binary random variables and consider the following tree graphical model representing the structure in the joint distribution p_{X_1,X_2,X_3,X_4}:



Moreover, suppose

$$\phi_{1,2}(x,x') = \phi_{1,3}(x,x') = \phi_{1,4}(x,x') = \begin{cases} 3 & x=0,\, x'=1 \\ 1 & \text{otherwise,} \end{cases}$$

$$\phi_2(x) = \phi_3(x) = \phi_4(x) = 1, \quad x \in \mathcal{X}, \quad \text{and} \quad \phi_1(x) = \begin{cases} \alpha & x=1 \\ 1 & x=0, \end{cases}$$

where $\alpha > 0$ is a constant. Compute all max-product algorithm messages and express the resulting most probable configuration (MPC) as a function of α.

Solution: We choose the node associated with X_1 as the root node and construct leaf-to-root messages:

$$m_{2\to1}(x_1) = \max_{x_2} \phi_{1,2}(x_1,x_2) = \begin{cases} 3 & x_1 = 0 \\ 1 & x_1 = 1, \end{cases}$$

and, by symmetry,

$$m_{3\to1}(x_1) = m_{4\to1}(x_1) = m_{2\to1}(x_1), \qquad x_1 \in \mathcal{X}.$$

In turn, the corresponding trace-back tables are

$$t_{1\to2}(x_1) = \arg\max_{x_2} \phi_{1,2}(x_1,x_2) = \begin{cases} 1 & x_1 = 0 \\ 0 \text{ or } 1 & x_1 = 1, \end{cases}$$

where both $x_2 = 0$ and $x_2 = 1$ achieve the maximum. By symmetry,

$$t_{1\to3}(x_1) = t_{1\to4}(x_1) = t_{1\to2}(x_1), \qquad x_1 \in \mathcal{X}.$$

The maximizing value of the root node variable is then

$$\hat{x}_1 = \arg\max_{x_1} g(x_1),$$

where

$$g(x_1) = \phi_1(x_1)\, m_{2\to1}(x_1)\, m_{3\to1}(x_1)\, m_{4\to1}(x_1) = \begin{cases} 1\cdot3\cdot3\cdot3 = 27 & x_1 = 0 \\ \alpha\cdot1\cdot1\cdot1 = \alpha & x_1 = 1. \end{cases}$$

Case $\alpha < 27$: In this case, $\hat{x}_1 = 0$ and we obtain the remaining values of the MPC via

$$\hat{x}_2 = t_{1\to2}(0) = 1,$$

and, once again by symmetry,

$$\hat{x}_3 = \hat{x}_4 = \hat{x}_2 = 1.$$

Hence, $(0, 1, 1, 1)$ is the (unique) MPC.

Case $\alpha > 27$: In this case, $\hat{x}_1 = 1$ and the trace-back tables imply that all configurations of the form $(1, x_2, x_3, x_4)$ are equally likely and are MPCs.

Case $\alpha = 27$: In this case, we obtain that $(0, 1, 1, 1)$ and all configurations of the form $(1, x_2, x_3, x_4)$ are equally likely and are MPCs.

Remark: Compare this solution to the sum-product result we obtained in an earlier recitation to see the difference in influence that α has on the most (marginally) likely value of X_1, as compared to the MPC.

Problem 10.3

Suppose you send your robot, Inquisitive, to Mars. Unfortunately, it gets stuck in a canyon while landing and most of its sensors break. You know the canyon has 3 areas. Areas 1 and 3 are sunny and hot, while Area 2 is cold. The temperature sensor still works. Every hour, you get a binary reading telling you whether the robot's current environment is `hot` or `cold`. The robot is equally likely to have landed in each of the three areas. Every hour, the robot tries to move forward by one area (i.e., from Area 1 to Area 2, or from Area 2 to Area 3). It succeeds with probability 3/4 and fails with probability 1/4. If it fails, it stays where it is. If it is in Area 3, it stays and waits to be rescued.

(a) Construct a hidden Markov model for this problem. Specify the initial state distribution, transition distribution, and the observation distribution.

Solution: We'll use 3 states, one per area.

The initial state distribution is uniform, i.e.,

$$p_{X_1}(x) = \frac{1}{3}, \qquad x \in \{1, 2, 3\}.$$

Based on the description above, we obtain the transition distribution $p_{X_{n+1}|X_n}$:

$p_{X_{n+1}\|x_n}(x_{n+1}\|x_n)$		x_{n+1}		
		1	2	3
x_n	1	1/4	3/4	0
	2	0	1/4	3/4
	3	0	0	1

Next, let's look at the observation distribution. There are two possible observations, hot and cold. Areas 1 and 3 always produce `hot` readings while Area 2 always produces a `cold` reading. Hence,

$$p_{Y_n|X_n}(\texttt{hot}|1) = p_{Y_n|X_n}(\texttt{hot}|3) = p_{Y_n|X_n}(\texttt{cold}|2) = 1.$$

(b) Suppose we observe the sequence (`hot`, `cold`, `hot`). Before doing any computation, determine the sequence of locations that produced these observations. Then compute the distribution of the second state using message passing. Do your answers match up?

Solution: Before doing any computation, we note that the sequence (`hot`, `cold`, `hot`) could only have been produced by the state sequence (1,2,3). Make sure you convince yourself this is true before continuing!

We start with the forward messages and compute

$$\begin{aligned}
m_{1\to 2}(x_2) &= \sum_{x_1} \tilde{\phi}_1(x_1)\, \phi_{1,2}(x_1, x_2) \\
&= \sum_{x_1} p_{x_1}(x_1)\, p_{Y_1|x_1}(\texttt{hot}|x_1)\, p_{X_2|X_1}(x_2|x_1) \\
&= p_{x_1}(1)\, p_{Y_1|x_1}(\texttt{hot}|1)\, p_{X_2|X_1}(x_2|1) \\
&\qquad + p_{x_1}(2)\, p_{Y_1|x_1}(\texttt{hot}|2)\, p_{X_2|X_1}(x_2|2) \\
&\qquad + p_{x_1}(3)\, p_{Y_1|x_1}(\texttt{hot}|3)\, p_{X_2|X_1}(x_2|3) \\
&= \frac{1}{3}\cdot 1 \cdot p_{X_2|X_1}(x_2|1) + \frac{1}{3}\cdot 0 \cdot p_{X_2|X_1}(x_2|2) + \frac{1}{3}\cdot 1 \cdot p_{X_2|X_1}(x_2|3) \\
&= \frac{1}{3}\, p_{X_2|X_1}(x_2|1) + \frac{1}{3}\, p_{X_2|X_1}(x_2|3),
\end{aligned}$$

or, in vector form,

$$m_{1\to 2} = \frac{1}{3}\cdot \begin{bmatrix} 1/4 \\ 3/4 \\ 0 \end{bmatrix} + \frac{1}{3}\cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{12}\cdot \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}.$$

The backwards messages are computed similarly:

$$\begin{aligned}
m_{3\to 2}(x_2) &= \sum_{x_3} \tilde{\phi}_3(x_3)\, \phi_{23}(x_2, x_3) \\
&= \sum_{x_3} p_{Y_3|x_3}(\texttt{hot}|x_3)\, p_{X_3|X_2}(x_3|x_2) \\
&= p_{Y_3|x_3}(\texttt{hot}|1)\, p_{X_3|X_2}(1|x_2) \\
&\qquad + p_{Y_3|x_3}(\texttt{hot}|2)\, p_{X_3|X_2}(2|x_2) \\
&\qquad + p_{Y_3|x_3}(\texttt{hot}|3)\, p_{X_3|X_2}(3|x_2) \\
&= p_{X_3|X_2}(1|x_2) + p_{X_3|X_2}(3|x_2),
\end{aligned}$$

or, in vector form,

$$m_{3\to 2} = \begin{bmatrix} 1/4 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 3/4 \\ 1 \end{bmatrix} = \frac{1}{4}\cdot \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

We can now compute the marginal distribution for X_2 given the data:

$$\begin{aligned} p_{X_2|Y_1,Y_2,Y_3}(x_2|\texttt{hot},\texttt{cold},\texttt{hot}) &\propto \tilde{\phi}_2(x_2)\, m_{1\to 2}(x_2)\, m_{3\to 2}(x_2) \\ &= p_{Y_2|X_2}(\texttt{hot}|x_2)\, m_{1\to 2}(x_2)\, m_{3\to 2}(x_2), \end{aligned}$$

or, in vector form,

$$p_{X_2|Y_1,Y_2,Y_3}(\cdot|\texttt{hot},\texttt{cold},\texttt{hot}) \propto \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \odot \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \odot \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 9 \\ 0 \end{bmatrix},$$

where $\odot$ denotes element-wise multiplication. Normalizing yields the final answer

$$p_{X_2|Y_1,Y_2,Y_3}(\cdot|\texttt{hot},\texttt{cold},\texttt{hot}) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Because of our simplified observation model, the observation `cold` enabled us to determine the state without any uncertainty. This matches up with our earlier conclusion that the robot must have been in Area 2 during the second hour.